

# Skills Summary

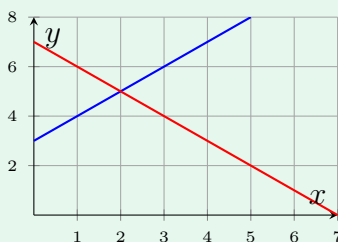
## Systems by Graphing

*Use this reference while completing your worksheet or when studying.*

### Skill 1 — Reading a crossing point off a graph

**Rule:** the crossing point lies on both lines, so its coordinates satisfy both equations. Read  $x$  across, then  $y$  up, and always substitute to check.

**Example:** Read the solution of  $y = x + 3$  and  $y = 7 - x$  from the graph.



**Step 1.** The lines cross at a grid corner, so read the coordinates there rather than solving — then confirm them in both equations.

**Step 2.** Two across and five up:  $5 = 2 + 3$  and  $5 = 7 - 2$ , both true.

On both lines at once. **(2, 5)**

**Try it:** Solve  $y = x + 1$  and  $y = 5 - x$ .

(**2**, **2**) :check

### Skill 2 — Graphing two lines from $y = mx + b$

**Rule:** plot the  $y$ -intercept, then step by the slope (right by the denominator, up by the numerator). Do this for both lines on one grid and read where they meet.

**Example:** Solve  $y = 2x + 1$  and  $y = x + 4$  by graphing.

**Step 1.** Both equations are already solved for  $y$ , so each one hands over a starting dot and a step — no rearranging is needed first.

**Step 2.** Start at 1 stepping up 2, and at 4 stepping up 1. The dots coincide at three across, seven up;  $2x + 1 = x + 4$  confirms  $x = 3$ .

Checked in both equations. **(3, 7)**

**Try it:** Solve  $y = 2x$  and  $y = x + 3$  by graphing.

(**3**, **6**) :check

### Skill 3 — Graphing a line given as $Ax + By = C$

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**Rule:** do not fight the form. Set  $x = 0$  to get one point, set  $y = 0$  to get another, draw the line through them — or solve for  $y$  first if the numbers are friendly.

**Example:** Solve  $x + y = 6$  and  $y = 2x$  by graphing.

**Step 1.** Only the first equation is in standard form, so get its two intercepts instead of rearranging it.

**Step 2.** Intercepts  $(0, 6)$  and  $(6, 0)$ ; the other line runs through the origin rising 2 each step. They meet where  $x + 2x = 6$ , so  $x = 2$ .

Two across, four up. **(2, 4)**

**Try it:** Solve  $x + y = 10$  and  $y = 4x$  by graphing.

check: (2, 8)

### Skill 4 — When the lines never meet

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**Rule:** equal slopes with different intercepts give parallel lines and *no solution*. Identical equations give one line drawn twice and *infinitely many* solutions. Say which, rather than guessing a point.

**Example:** Solve  $y = 2x + 1$  and  $y = 2x + 5$  by graphing.

**Step 1.** Compare the slopes before drawing anything: both are 2, so the lines can only be parallel or identical.

**Step 2.** The intercepts differ, 1 against 5, so the lines stay 4 units apart and never cross.

Nothing lies on both lines. **no solution**

**Try it:** Solve  $y = 4x + 2$  and  $y = 4x + 7$  by graphing.

check: no solution